

# Estimating spatial firm productivity spillovers using UK microdata

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## DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistative role (UCL Category 2) for this submission. I used Gemini 3.10 Pro (DeepMind, 2026). This is limited to the roles of: supporting the particular process of reviewing and refactoring code (data analysis), as a support tutor (model and empirical approach).

Name: Aloysius Lip

Date: 2nd September, 2026

## CLASSIFICATION

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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## TABLE OF CONTENTS

|     |                                                                      |    |
|-----|----------------------------------------------------------------------|----|
| 1   | Introduction . . . . .                                               | 1  |
| 2   | Literature . . . . .                                                 | 1  |
| 3   | Data . . . . .                                                       | 2  |
| 3.1 | Fame . . . . .                                                       | 2  |
| 3.2 | Ingestion and cleaning . . . . .                                     | 2  |
| 3.3 | Location data . . . . .                                              | 2  |
| 3.4 | Deriving panel aggregates . . . . .                                  | 2  |
| 3.5 | Firm distance measures . . . . .                                     | 3  |
| 4   | Static models . . . . .                                              | 5  |
| 4.1 | 2-factor TFP estimation . . . . .                                    | 5  |
| 4.2 | Static panel identification . . . . .                                | 6  |
| 4.3 | LMM grouped peers . . . . .                                          | 8  |
| 4.4 | Static distance decay . . . . .                                      | 10 |
| 5   | Dynamic models . . . . .                                             | 11 |
| 5.1 | Instrumental identification under no error autocorrelation . . . . . | 11 |
| 5.2 | Autoregressive specification . . . . .                               | 12 |
| 5.3 | Instrumental identification under error autocorrelation . . . . .    | 16 |
| 5.4 | Panel vector autoregression . . . . .                                | 16 |
| 5.5 | Alternative weights on peer TFP . . . . .                            | 16 |
|     | Bibliography . . . . .                                               | 17 |
|     | Appendix 1 . . . . .                                                 | 18 |
|     | Appendix 2 . . . . .                                                 | 22 |
|     | Appendix 3 . . . . .                                                 | 23 |

## LIST OF FIGURES

## LIST OF TABLES

|      |                                                                                |    |
|------|--------------------------------------------------------------------------------|----|
| I    | Capital-output ratios in FAME corpus of firms . . . . .                        | 3  |
| II   | Baseline TFP calculation from 2-factor production . . . . .                    | 6  |
| III  | Static panel linear-in-means using spatial-group TFP averages . . . . .        | 9  |
| IV   | Static panel using continuous distance-decay weighted TFP average of peers . . | 10 |
| V    | Dynamic panel: all instrumented variables . . . . .                            | 14 |
| VI   | Dynamic panel using instrumented continuous distance-decay weighted TFP . .    | 15 |
| VII  | Dynamic panel: higher-order autoregressive lags . . . . .                      | 18 |
| VIII | Dynamic panel: higher-order regressor lags 1-5 . . . . .                       | 19 |
| IX   | Dynamic panel: higher-order regressor lags 6-10 . . . . .                      | 20 |
| X    | Dynamic panel: higher-order VAR(2,4) structures . . . . .                      | 21 |

## ABSTRACT

~300 word summary of your research topic, methodology and findings

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Word count:

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## 1. INTRODUCTION

## 2. LITERATURE

My investigation is rooted in Marshall (1920)’s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers. There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman also labels all these channels as ‘centripetal’ to an urban core; there are likewise ‘centrifugal’ factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have “increasing strength” attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently (Bloom, Schankerman and Van Reenen, 2013).

Knowledge spillovers have been a well-evidence factor of within-firm managerial-derived productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

### 3. DATA

#### 3.1. FAME

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular, number of employees and balance sheet data from which analysis-relevant aggregates can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey<sup>5</sup>.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim for (Irastorza-Fadrique et al., 2026). and reused for this project. Raw data is provided on a series of a financial balance sheet indicators, including characteristics that are fixed for each firm, bucketed into *a1\_ID* and *a5\_misc* indicator tables, and annual data published on a calendar-year basis, bucketed into *a2\_key\_finance*, *a3\_assets*, and *a4\_profits* tables. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

#### 3.2. INGESTION AND CLEANING

(...)

All data is deflated to real terms, using 2024 as the base year. I used the GDP deflator for all balance sheet variables, except for *renumeration\_employees* and its composites for which I used CPIH, as these are income-related aggregates with real spending power best captured by household purchasing baskets<sup>6</sup>.

All balance sheet data are in thousands ('000s) of GBP (£). As we estimate log relationships exclusively, this has no effect on parameter interpretation.

#### 3.3. LOCATION DATA

#### 3.4. DERIVING PANEL AGGREGATES

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame's *fixed\_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values Table (I), which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

GVA is calculated as the sum of *ebitda* and *renumeration\_employees*. *renumeration\_employees*

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<sup>5</sup>Analytics (2025)

<sup>6</sup>for National Statistics (2026)

TABLE I  
Capital-output ratios in FAME corpus of firms

| Variable            | Description                                               | Capital-output ratios |
|---------------------|-----------------------------------------------------------|-----------------------|
| <i>tangibles</i>    |                                                           | 3.53                  |
| <i>fixed_total</i>  | <i>tangibles</i> + <i>intangibles</i> + <i>inv._other</i> | 6.00                  |
| <i>total_assets</i> | <i>fixed_total</i> + <i>current_assets</i>                | 11.56                 |

is the exact sum of *wages*, *social\_security\_costs*, and *pensions\_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via the sum of *profit\_loss\_pretax*, *interest\_paid*, and *depreciation*, providing an alternative measure of GVA. We can expect the difference between these two to be accounting differences<sup>7</sup> and additionally note that the alternate GVA measure also reduces panel observations from 1,085,520 to 618,588. For these reasons, I do not report TFP calculations with this second GVA measure which are regardless near-identical, though I resultingly refer to GVA throughout this paper as *gva\_1*.

### 3.5. FIRM DISTANCE MEASURES

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. This relationship is irreflexive. For groups it is transitive: if firms A and B share a spatial group, such as a travel-to-work area (*ttwa*), first-half of a UK postcode (*pc4*), or full postcode (*pc8*), then a firm C that is a peer of firm B must also be of firm A. For continuous distance measures, this is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are not shared with A. Every firm is then its own overlapping group, or equivalently, a sparse, rather than block diagonal, firm-to-firm matrix<sup>8</sup>.

#### (Cleaning address data).

Fame only reports each company’s last-known registered and trading address, at the time of compilation of publication of its annual vintage(cite). As a result, final location data is only available at the firm level. We can interpret each firm as being static over the entire 20-year sample, with no movers. This seems very unlikely (expand). Instead, we must interpret the constructing address, and therefore firm-to-firm distance measures as a signal of firm pairs’ true distance polluted by movers. This has 2 econometric consequences.

Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, their firm location decision is endogenous to the subsequent productivity spillovers they experience and exert, for reasons discussed in Section 2 on New Economic Geography. (More on firm FEs) There is no solution to this for moves occurring out-of-sample, for instance, just before 2009.

Secondly, I acknowledge some attenuation to the productivity spillover estimate Peer firms who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location (More on direction of bias/positive

<sup>7</sup>An OLS regression between the two, discarding 5% outliers, has a coefficient of 0.91 with a 0.000 p-value. This means *gva2* is slightly lower than *gva1*, with that impact mostly driven by top-end outliers.

<sup>8</sup>I do actually impose a block diagonal structure for reasons of computation as explained subsequently

selection?)

I first consider a model where each target has peers which share a spatial identifier, meaning they belong to the same travel-to-work-area (*ttwa*) using the 2011 definitions <sup>9</sup>, the first half of a UK postcode *pc4*, and the full UK postcode *pc8*.

(Discuss and illustrate different forms of  $\omega$ , with illustrative block diagonal matrices)

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<sup>9</sup>(Review ONS bulletin on TTWAs data source and describe clearly)

#### 4. STATIC MODELS

I use a stylised 3-factor production model throughout this analysis.  $Y$ ,  $A$ ,  $K$ , and  $L$  are each firm-year observation's structural GVA, TFP, labour, and capital as usual. I use lower-case variables to indicate logged terms in reduced form and bolded variables to indicate stacked vectors.

Log TFP, denoted by  $z_{i,t}$ , is the reduced-form residual after controlling for the two classical inputs, which are assumed to be exogenous and exhibit unit factor substitutability in this static model. Under this specification, the residual is equivalent to the sum of a constant across all observations, firm and time fixed effects, and most crucially, the weighted sum of all peer log TFP  $\sum_{j \neq i} w_{ij} z_{j,t}$ , with a structural  $\delta_0$  parameter to estimate.

$$Y_{i,t} = A_{i,t} \cdot K_{i,t}^{\beta_1} \cdot L_{i,t}^{\beta_2} \quad (1)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (2)$$

$$z_{i,t} = \ln(A_{i,t}) = y_{i,t} - \alpha_i - \beta_1 k_{i,t} - \beta_2 l_{i,t} \quad (3)$$

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (4)$$

Theory suggests no restrictions on  $\beta_1$ ,  $\beta_2$ , and  $\delta_0$  collectively should be imposed, as  $\delta_0$  is an externality. (Lucas discusses this)

The reduced-form equation 2 is linear in the weights  $w_{ij}$  applied to peer log TFP and its coefficient. This permits me to estimate  $\delta_0$  through either the full GVA specification in one stage, using the log of *gva1* as the dependent variable, or in two stages, calculating  $z_{i,t}$  as the 2-factor residual and then regressing  $z_{i,t}$  however desired in reduced form. This is convenient for a number of the static models below.

##### 4.1. 2-FACTOR TFP ESTIMATION

For convenience, I initially conduct the 2-factor estimation deriving the residual  $z_{i,t}$  to decompose further in a second stage. This step is additionally helpful for validating the relationships about the derived aggregates from my data.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + u_{i,t} \quad (5)$$

$$u_{i,t} = \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t}$$

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + u_{i,t} \quad (6)$$

$$\hat{z}_{i,t} = y_{i,t} - \hat{\beta}_1 k_{i,t} - \hat{\beta}_2 l_{i,t} \quad (7)$$

I conduct the estimation using the log-transformed variables. I must include firm and time fixed effects and a constant if equation (4) is the true specification, otherwise the static panel OLS will inaccurately attribute firm and time variation to my classical factor inputs (explain more clearly). As outlined above, this analysis assumes factor input exogeneity.

The parameter coefficients give a labour income share exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share*

TABLE II  
Baseline TFP calculation from 2-factor production

|                               | (1)                     | (2)                     | (3)                     | (4)                     | (5)                     |
|-------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|                               | <i>gva</i> <sub>1</sub> | <i>gva</i> <sub>1</sub> | <i>gva</i> <sub>1</sub> | <i>gva</i> <sub>1</sub> | <i>gva</i> <sub>1</sub> |
| constant                      | 3.201<br>(0.032)***     | 4.840<br>(0.019)***     | 3.223<br>(0.027)***     | 1.994<br>(0.010)***     | 1.996<br>(0.010)***     |
| $\ln(\textit{employees})$     | 0.621<br>(0.005)***     | 0.717<br>(0.005)***     | 0.618<br>(0.004)***     | 0.559<br>(0.002)***     | 0.559<br>(0.002)***     |
| $\ln(\textit{fixed\_total})$  |                         | 0.058<br>(0.001)***     |                         |                         |                         |
| $\ln(\textit{total\_assets})$ | 0.265<br>(0.004)***     |                         | 0.264<br>(0.004)***     | 0.422<br>(0.002)***     | 0.422<br>(0.002)***     |
| <i>i</i> fixed effects        | ✓                       | ✓                       | ✓                       |                         |                         |
| <i>t</i> fixed effects        | ✓                       | ✓                       |                         | ✓                       |                         |
| Observations                  | 1,081,520               | 1,042,854               | 1,081,520               | 1,081,520               | 1,081,520               |
| R <sup>2</sup>                | 0.3701                  | 0.3226                  | 0.3861                  | 0.8345                  | 0.8346                  |

All parameters reported with standard errors underneath. Significant estimates against an  $H_0 : \beta = 0$  are indicated with  $p < 0.1$  (\*),  $p < 0.05$  (\*\*), and  $p < 0.01$  (\*\*\*).

which is constructed from a similar sample to the coverage in Fame<sup>10</sup>.

Surprisingly, running the regression with *fixed\_total* as the capital measure instead of *total\_assets* almost entirely eliminates the capital share of income, or explanatory power of capital on variation in GVA. I leave it for future analysis why this dataset yields this phenomenon and for the rest of the analysis, I use *total\_assets* as the primary capital measure.

As a demonstration, I include specifications which omit fixed effects. The capital share of income jumps on a specification with only time fixed effects, which is to be expected (explain).

TFP is the residual on GVA after extracting the proportion that can be accounted by capital and labour inputs. Equivalently, it is the sum of fixed effects in this specification with the error term, which means we should expect it to be exogenous to capital and labour inputs, but correlated to both firm effects and time trends. By extracting this regression residual, I have a TFP measure for all of my 1,081,520 panel observations.

#### 4.2. STATIC PANEL IDENTIFICATION

Models in this section are consistent only if the regressors are plausibly exogenous from  $\varepsilon_{i,t}$ . Following standard procedure for a wide panel regression, meaning many firm observations and far fewer time periods, I transform all variables to first differences.

<sup>10</sup>for National Statistics (2024)

$$\begin{aligned}
z_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \\
&= \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_{j \neq i} w_{ij} \cdot (\alpha_j + \gamma_t + \beta_0 + \delta_0 \sum_{k \neq j} w_{jk} z_{k,t} + \varepsilon_{j,t}) + \varepsilon_{i,t} \tag{8}
\end{aligned}$$

$$\Delta z_{i,t} = \Delta \gamma_t + \delta_0 \sum_{j \neq i} w_{ij} \cdot (\Delta \gamma_t + \delta_0 \sum_{k \neq j} w_{jk} \Delta z_{k,t} + \Delta \varepsilon_{j,t}) + \Delta \varepsilon_{i,t} \tag{9}$$

$$\begin{aligned}
&= \Delta \gamma_t \cdot (1 + \delta_0 \sum_{j \neq i} w_{ij}) + \delta_0^2 \sum_{j \neq i} \sum_{k \neq j} w_{ij} w_{jk} \Delta z_{k,t} + \underbrace{\delta_0 \sum_{j \neq i} \Delta \varepsilon_{j,t} + \Delta \varepsilon_{i,t}}_{\nu_{i,t}} \tag{10}
\end{aligned}$$

For these models, I make the strong assumption of no structural, cross-sectional correlation between error terms. This means that for any productivity shock  $\varepsilon_{j,t}$  that hits a peer firm  $j$ , if it is correlated with a productivity shock  $\varepsilon_{i,t}$  occurring in the target firm  $i$  within the same period  $t$ , it must be attributable to the peer effect coefficient  $\delta$ .

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] = 0 \tag{11}$$

This is quite a strong assumption. It precludes any contemporaneous, correlated independent TFP shocks, which might occur on both the global and local level. (...)

#### 4.3. LMM GROUPED PEERS

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (12)$$

$$\mathbf{z}_t = \alpha + \gamma_t \iota + \beta_0 \iota + \delta_0 \mathbf{W} \mathbf{z}_t + \varepsilon_t \quad (13)$$

$$= (\mathbb{I} - \delta_0 \mathbf{W})^{-1} \cdot (\alpha + \gamma_t \iota + \beta_0 \iota + \varepsilon_t) \quad (14)$$

$$= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \cdot (\alpha + \gamma_t \iota + \beta_0 \iota + \varepsilon_t) \quad (15)$$

Section 3.5 discusses how firms have been labelled to symmetric, transitive, non-overlapping groups by *pc8* (smallest), *pc4*, and *twa* (largest). I conduct a linear regression using the averages of peers belonging to these groups, leaving out the target firm. This is known as the 'Linear-in-Means (LMM)' Model and is common to the peer effects literature.

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \sum_{g=1}^3 \beta_g \mathbb{E}[z_{itg}|g] + \varepsilon_{it} \quad (16)$$

$$\begin{aligned} \mathbf{z}_t &= \delta \mathbf{W} \mathbf{z}_t \\ w_{ij} \in \mathbf{W} &= \begin{cases} 1, & \text{if } (i, j) \in g \\ 0, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases} \end{aligned} \quad (17)$$

It is crucial that I have multiple, non-overlapping groups in a single model, where my target firm  $z_{it}$  is a member of all groups simultaneously, but peers in one group are excluded from all others.

The non-overlapping property prevents multicollinearity. If I were to use nested, overlapping geographies, namely keeping all firms within the *twa* area in that group average, including the ones I have just counted in the *pc4* and *itxpc8* groups, then my group averages mechanically align:  $z_{jt,pc8}$  is embedded within  $z_{jt,pc4}$ , and both are within  $z_{jt,twa}$ , making  $\beta_g$  parameter estimates unstable.

The single model approach prevents omitted variable bias (OVB).  $\mathbb{E}[z_{jt,twa}|g] \propto \mathbb{E}[z_{jt,pc8}|g]$ , so running three models where the former is just my single regressor means the latent  $\beta_{pc8} \in u_{it}$  is correlated with both  $z_{itg}$  and  $z_{jtg}$ . We could reasonably argue that the inner rings are causal, so this causal factor excluded from the regression influences both and creates heavily upward bias on my outer-ring estimations. Indeed, this critique extends within further inner rings. If I think the causal unit is not peer firms within the *pc8* group at all, but a further hyper-local effect - perhaps those on the same street or same building, then this latent factor will correlate with both my outcome and regressor and drive the results. However, given the small number of peers contained within an *pc8* unit and the limits of data availability to identify firms at more local granularities, I proceed assuming the effect from *pc8* peers is homogenous.

Finally, the approach using multiple groups as regressors at once gives identification by breaking transitivity. If I were to run the same regression with merely one group, say *pc8*, then my regression would suffer the reflection problem outlined in (Manski, 1993). As I am estimating the expected value of  $z_{it}$  and  $z_{jt}$  from the same sample, they cannot be uniquely identified in reduced form.

*Reflection problem with a single regressor,  $\iota$  denotes a column vector of 1s*

$$\begin{aligned}
\mathbb{E}[z_{itg}|g] &= \alpha_i + \gamma_t + \beta_0 + \beta_1 \mathbb{E}[z_{jtg}|g] \\
&= (\alpha_i + \gamma_t + \beta_0)\iota + \beta_1 \mathbf{\Omega}_t \mathbb{E}[\mathbf{z}_{tg}|g] \\
&= (\mathbb{I} - \beta_1 \mathbf{\Omega}_t)^{-1}(\alpha_i + \gamma_t + \beta_0)\iota
\end{aligned}$$

This problem is mitigated with having 3 distinct groups to which  $z_{it}$  belongs. The results of the LMM estimation implementing equation 16 are below:

TABLE III  
Static panel linear-in-means using spatial-group TFP averages

|                                 | (1)<br><i>tfp</i>   | (2)<br><i>tfp</i>   | (3)<br><i>tfp</i>    |
|---------------------------------|---------------------|---------------------|----------------------|
| constant                        | 2.075<br>(0.044)*** | 0.310<br>(0.038)*** | 1.202<br>(0.215)***  |
| $\ln(\text{employees})$         |                     |                     | 0.184<br>(0.044)***  |
| <i>peer_tfp_pc8</i>             | 0.280<br>(0.005)*** | 0.559<br>(0.004)*** | 0.280<br>(0.005)***  |
| <i>peer_tfp_pc4_donut</i>       | 0.015<br>(0.005)*** | 0.142<br>(0.008)*** | 0.015<br>(0.005)***  |
| <i>peer_tfp_ttwa_donut</i>      | 0.033<br>(0.013)*** | 0.204<br>(0.013)*** | 0.306<br>(0.071)***  |
| <i>peer_tfp_pc8_x_employees</i> |                     |                     | -0.058<br>(0.015)*** |
| <i>i</i> fixed effects          | ✓                   |                     | ✓                    |
| <i>t</i> fixed effects          | ✓                   | ✓                   | ✓                    |
| Observations                    | 764,537             | 764,537             | 764,537              |
| R <sup>2</sup>                  | 0.0388              | 0.1792              | 0.0390               |

The parameter results suggest a strong correlation between the TFP of a target firm and its peers.

#### 4.4. STATIC DISTANCE DECAY

To improve on the LMM specification, I make a modification where every peer firm is weighted by its influence on the target by the reciprocal of the continuous distance metric calculated in 3.5  $w_{ijt} = f(d)$ . This is stylistic of a declining peer influence further away, representing a (lower frequency of social collisions, barriers created via transport and time costs, and an increased difficulty of observation of management practices).

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \beta_1 \sum_{j \neq i} w_{ijt} z_{jt} + \varepsilon_{it} \quad (18)$$

$$\mathbf{z}_t = (\alpha_i + \gamma_t + \beta_0) \iota + \beta_1 \sum_{j \neq i} \Omega_t \mathbf{z}_t + \varepsilon_t \quad (19)$$

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2} & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (20)$$

This represents a likely more accurate declining effect of productivity spillover, rather than the special-case group mean where every firm in an enclosed area, for example, the target's postcode, has an equal influence on the target's TFP. In addition, it allows for better identification following Bramoullé et al. (2009). (explain Bramouille2009, patterns of ) $\Omega_t$  In essence, the LMM specification only contains 3 non-overlapping groups from its location tags. The distance-decayed specification contains as many non-overlapping groups as firms.

TABLE IV  
Static panel using continuous distance-decay weighted  
TFP average of peers

|                             | (1)<br><i>tfp</i>   | (2)<br><i>tfp</i>   | (3)<br><i>tfp</i>   |
|-----------------------------|---------------------|---------------------|---------------------|
| constant                    | 2.479<br>(0.029)*** | 2.586<br>(0.029)*** | 2.417<br>(0.032)*** |
| $\ln(\textit{employees})$   | -0.009<br>(0.004)** | -0.009<br>(0.004)** | -0.009<br>(0.004)** |
| $\ln(\textit{nb\_peers})$   | 0.013<br>(0.004)*** | 0.012<br>(0.004)*** | 0.016<br>(0.004)*** |
| <i>tfp_wav</i> <sub>1</sub> | 0.168<br>(0.003)*** |                     |                     |
| <i>tfp_wav</i> <sub>2</sub> |                     | 0.135<br>(0.003)*** |                     |
| <i>tfp_wav</i> <sub>3</sub> |                     |                     | 0.184<br>(0.005)*** |
| <i>i</i> fixed effects      | ✓                   | ✓                   | ✓                   |
| <i>t</i> fixed effects      | ✓                   | ✓                   | ✓                   |
| Observations                | 1,067,013           | 1,067,013           | 1,067,013           |
| R <sup>2</sup>              | 0.0101              | 0.0095              | 0.0040              |

This distance-decay specification revises down the estimate from peers to a an elasticity of 0.168, lower than the results given by the LMM specification. This is likely due to (...) In addition, I introduce controls (...)

## 5. DYNAMIC MODELS

I have discussed how the static model requires a strong strict exogeneity assumption. I showed in subsection 4.2 that cross-sectional orthogonality is a **(necessary condition)** to meet this assumption with a peer-effects regressor.

$$\begin{aligned}
 y_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \\
 \Delta y_{i,t} &= \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \\
 \mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] &= 0, \quad \forall t \\
 \implies \mathbb{E}[\varepsilon_{i,t} | x_{i,t}] &= 0, \quad \forall t \quad x_{i,t} = \begin{pmatrix} k_{i,t} & l_{i,t} & \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix} \\
 \implies \mathbb{E}[\Delta \varepsilon_{i,t} | \Delta x_{i,t}] &= 0, \quad \forall t
 \end{aligned}$$

In this section, I introduce a dynamic panel specification to relax these assumptions, permitting weak cross-sectional correlation between peer errors and endogeneity of the contemporaneous regressors. Crucially, this specification can still be identified even in its panel reduced form, which requires first differences to eliminate fixed effects, by the use of internal instruments. This follows econometric theory developed by (Anderson and Hsiao, 1981) and estimator techniques developed in (Arellano and Bond, 1991) and furthered by (Blundell and Bond, 1998) which improve efficiency.

There are good reasons to think this weak cross-sectional correlation exists. A local demand shock **(...)**

### 5.1. INSTRUMENTAL IDENTIFICATION UNDER NO ERROR AUTOCORRELATION

The estimator given in (Arellano and Bond, 1991) takes the regression equation as given above and allows for endogeneity. Internal instruments of all the endogenous regressors are found, meaning their lagged first differences. Identification is improved in the System GMM approach given in (Blundell and Bond, 1998) by also including the first lagged, not differenced, lag of the endogenous regressors which meets the usual instrument conditions, as an instrument. It is this System GMM approach that I follow here, so I find the first lagged, not differenced, lag of regressor in  $x_{it}$  and its corresponding set of first difference lagged instruments, which in practice just start from  $\Delta x_{i,\tau-j}$  for any  $j$  if  $x_{i,\tau}$  is a valid instrument.

I commence finding these valid instruments by assuming the structural specification (2) exhibits no autocorrelation in its error term  $\varepsilon_{i,t}$ , though I now allow for moderate cross-sectional correlation across firms.

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{i,t-1}] = 0, \quad \forall t, t-1 \quad (21)$$

$$\mathbb{E}[x_{i,t} \varepsilon_{i,t}] \neq 0, \quad \exists i \quad (22)$$

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] \neq 0, \quad \exists i, j \quad (23)$$

$$\mathbb{E}[\varepsilon_t] = 0, \mathbb{E}[\varepsilon_t^2] = \sigma_t^2, \mathbb{E}[\varepsilon_t | \Omega_{t-1}] = 0, \quad \forall i \quad (\text{w.n.}) \quad (24)$$

$$\begin{aligned}
 x_{i,t} &= \begin{pmatrix} k_{i,t} & l_{i,t} & \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix} \\
 \Delta x_{i,t} &= \begin{pmatrix} \Delta k_{i,t} & \Delta l_{i,t} & \Delta \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix}
 \end{aligned}$$

Taking each regressor in turn, I relax the exogeneity assumption of the standard factors in the reduced-form equation,  $\Delta k_{i,t}$  and  $\Delta l_{i,t}$ . I assume a new optimal firm decision on capital

based on information, denoted by  $\Omega_t$  which includes  $\varepsilon_{i,t}$  for its respective  $t$ . This approach allows the discovery of valid instruments which are exogenous to  $\varepsilon_{i,t}$  and  $\Delta\varepsilon_{i,t}$ . I justify an exclusion restriction for  $\Delta k_{i,t-j} \forall j > 1$ , as it is predetermined, and  $\Delta l_{i,t-k}, k > 2$ , as it is flexible.

$$K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (25)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* \mid \Omega_{t-1}] \quad (26)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* \mid \Omega_t] \quad (27)$$

$$\varepsilon_{i,t} \in \Omega_t$$

$$\implies K_{i,t} \propto \varepsilon_t, \varepsilon_{t-1}$$

$$\implies L_{i,t} \propto \varepsilon_t$$

$$\Delta\varepsilon_{i,t} = \varepsilon_{i,t} - \varepsilon_{i,t-1}$$

$$\Delta k_{i,t} = k_{i,t} - k_{i,t-1} \propto \varepsilon_{i,t}, \varepsilon_{i,t-1}, \varepsilon_{i,t-2}$$

$$\implies \mathbb{E}[\Delta k_{i,t}, \Delta\varepsilon_t] \neq 0$$

$$\implies \mathbb{E}[\Delta k_{i,t-1}, \Delta\varepsilon_t] = 0 \quad (28)$$

$$\Delta l_{i,t} = l_{i,t} - l_{i,t-1} \propto \varepsilon_{i,t}, \varepsilon_{i,t-1}$$

$$\implies \mathbb{E}[\Delta l_{i,t-2}, \Delta\varepsilon_t] = 0 \quad (29)$$

## 5.2. AUTOREGRESSIVE SPECIFICATION

A dynamic specification grants the flexibility to add past realisations of variables to the decomposition of TFP,  $z_{i,t}$ . This increases the number of regressors in reduced form, improving the plausibility of its exogeneity with  $\varepsilon_{i,t}$ .

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \phi_1 y_{i,t-1} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (30)$$

The above equation is in structural form in the 1-stage regression, but I can find an equivalent in the 2-stage regression with  $z_{i,t}$  as the dependent variable.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \phi_1 y_{i,t-1} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (31)$$

$$= \alpha_i + \gamma_t + \beta_0 + \phi_1 \cdot (\beta_1 k_{i,t-1} + \beta_2 l_{i,t-1} + z_{i,t-1}) + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (32)$$

For system GMM, I need to show that the reduced-form first difference regressors are exogenous.

$$\Delta y_{i,t} = \Delta\gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \sum_{k=1}^p \phi_k \Delta y_{i,t-k} + \delta_0 \sum_{j \neq i} w_{ij,t} \Delta z_{j,t} + \Delta\varepsilon_{i,t} \quad (33)$$

$$\Delta z_{i,t} = \Delta\gamma_t + \phi_1 \cdot (\beta_1 \Delta k_{i,t-1} + \beta_2 \Delta l_{i,t-1} + \Delta z_{i,t-1}) + \delta_0 \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta\varepsilon_{i,t} \quad (34)$$

The first-order autoregressor,  $\Delta \ln(y_{it})$  is endogenous.

$$\begin{aligned}
\Delta \varepsilon_{i,t} &= \varepsilon_{i,t} - \varepsilon_{i,t-1} \\
\Delta y_{i,t-1} &= \dots + \Delta \varepsilon_{i,t-1} \\
&= \dots + \varepsilon_{i,t-1} - \varepsilon_{i,t-2} \\
&\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] \neq 0 \\
\Delta y_{i,t-2} &= \dots + \varepsilon_{i,t-2} - \varepsilon_{i,t-3} \\
&\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] = 0
\end{aligned} \tag{35}$$

This is confirmed by the Anderson-Rubin (AR(1)) test in my results, which rejects the null hypothesis at the 1% level for all models, suggesting that first-order lags of log GVA are a weak instrument (Why? Shouldn't they fail exclusion restriction, not instrument correlation). In addition, I assume that capital,  $k_{i,t}$  is pre-determined, as investment is chosen one period ahead, in line with standard Perpetual Inventory Method approaches<sup>11</sup> (Review predetermined material. Additionally, explain the logic of instrumenting for capital)

To overcome this endogeneity, I introduce deeper autoregressive lags as instrumental variables, as proposed by (Anderson and Hsiao, 1981). I use 3 autoregressive instrument lags starting from  $t-2$ :  $\ln(y_{i,t-2}), \ln(y_{i,t-3}), \ln(y_{i,t-p-3})$  for log GVA, and 2 lags for capital  $\ln(k_{i,t-p-1}), \ln(k_{i,t-p-2})$ . An Anderson-Rubin (AR(2)) test shows that deep lags are a valid instrument for log GVA. The test applied to the second lag of  $\ln(gva_1)$  fails to reject the null hypothesis of for any specification, so I assume it is a valid instrument.

Additionally, using multiple deep lags allows for overidentification: in this basic AR(1) specification, I have  $k = 4$  regressors and  $l = 7$  instruments. I could increase this. Fame is rich on the time dimension, with up to 19 lags at most for a firm, and a median of 6 lags, so I could safely provide more lags of regressors as instruments. Further lags is likely to have a decreasing marginal effect on explanatory power and additionally ((...))

(Discuss introducing a new weighting method using past firm similarity.)

I estimate the equation fulfilling the vector of moment conditions, (36) via GMM.

$$\mathbb{E} \left[ \Delta \varepsilon_{it} \cdot \begin{pmatrix} \Delta \ln(y_{i,t-2}) \\ \Delta \ln(y_{i,t-3}) \\ \Delta \ln(y_{i,t-4}) \\ \Delta \ln(k_{i,t-2}) \\ \Delta \ln(k_{i,t-3}) \\ \Delta \Sigma_j w_{ijt} z_{jt} \end{pmatrix} \right] = 0 \tag{36}$$

As my regressors are over-identified, I use 2-step GMM to improve efficiency. (I should introduce a control (basically a diff-in-diff?) of peers from the same industry across the country (outside the spatial bubble).)

Finally, a Sargan-Hansen J-test is implemented to check if these instruments are over-identified, meaning the moment conditions cannot be fulfilled with 95% confidence. In all models, the J-test fails to reject the null hypothesis, meaning the model's moment conditions are jointly correct.

<sup>11</sup>Noting that this is ordinarily done on a quarterly basis, however, due to data limitations we adopt the same thing here even though Fame provides observations annually.

TABLE V  
Dynamic panel: all instrumented variables

|                      | (1)<br>$\ln(gva_1)$ | (2)<br>$\ln(gva_1)$ | (3)<br>$\ln(gva_1)$ |
|----------------------|---------------------|---------------------|---------------------|
| constant             | 0.653<br>(0.345)    | 0.031<br>(0.215)    | -0.656<br>(0.970)   |
| $\ln(gva_1)_{t-1}$   | 0.006<br>(0.001)*** | 0.003<br>(0.001)*** | 0.004<br>(0.002)    |
| $tfp\_wav_{1t-1}$    |                     |                     | -0.017<br>(0.012)   |
| $\ln(employees)$     | 0.523<br>(0.051)*** | 0.655<br>(0.027)*** | 0.599<br>(0.075)*** |
| $\ln(total\_assets)$ | 0.310<br>(0.030)*** | 0.298<br>(0.030)*** | 0.283<br>(0.036)*** |
| $tfp\_wav_1$         | 0.832<br>(0.087)*** | 0.887<br>(0.081)*** | 1.251<br>(0.303)*** |
| Hansen Test          | ✓<br>(0.049)**      | ✓<br>(0.015)**      | ✗<br>(0.941)        |
| AR(1) Test           | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     |
| AR(2) Test           | ✗<br>(0.119)        | ✗<br>(0.879)        | ✗<br>(0.117)        |
| Instruments          | 35                  | 35                  | 52                  |
| $i$ fixed effects    | ✓                   | ✓                   | ✓                   |
| $t$ fixed effects    | ✓                   | ✓                   | ✓                   |
| Observations         | 713,339             | 713,339             | 713,339             |

Model includes a constant as the baseline for  $t$  fixed effects.

TABLE VI  
Dynamic panel using instrumented continuous  
distance-decay weighted TFP

|                      | (1)<br>$\ln(gva_1)$ | (2)<br>$\ln(gva_1)$ | (3)<br>$\ln(gva_1)$ |
|----------------------|---------------------|---------------------|---------------------|
| constant             | 1.371<br>(0.134)*** | 1.577<br>(0.144)*** | 1.064<br>(0.103)*** |
| $\ln(gva_1)_{t-1}$   | 0.005<br>(0.001)*** | 0.005<br>(0.001)*** | 0.005<br>(0.001)*** |
| $\ln(employees)$     | 0.696<br>(0.038)*** | 0.689<br>(0.037)*** | 0.709<br>(0.040)*** |
| $\ln(total\_assets)$ | 0.259<br>(0.041)*** | 0.266<br>(0.041)*** | 0.247<br>(0.044)*** |
| $tfp\_wav_1$         | 0.503<br>(0.028)*** |                     |                     |
| $tfp\_wav_2$         |                     | 0.423<br>(0.023)*** |                     |
| $tfp\_wav_3$         |                     |                     | 0.618<br>(0.042)*** |
| Hansen Test          | X<br>(0.417)        | X<br>(0.459)        | X<br>(0.246)        |
| AR(1) Test           | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     |
| AR(2) Test           | X<br>(0.746)        | X<br>(0.600)        | X<br>(0.945)        |
| $i$ fixed effects    | ✓                   | ✓                   | ✓                   |
| $t$ fixed effects    | ✓                   | ✓                   | ✓                   |
| Observations         | 713,339             | 713,339             | 713,339             |

Model includes a constant as the baseline for  $t$  fixed effects.

### 5.3. INSTRUMENTAL IDENTIFICATION UNDER ERROR AUTOCORRELATION

I appear to have strong identification given assumptions (21) and (23). The instruments above exhibit strong correlation under Anderson-Rubin tests and exclusion restriction has been derived. However, there exists a niche case of diagonally-correlated errors, namely despite (21) and (23), I ought to consider the following:

$$\mathbb{E}[\varepsilon_{i,t}\varepsilon_{j,t-k}] \neq 0, \quad \exists i, j, k \quad (37)$$

This is not as implausible as it may appear due to this analysis' peer effects setting. There may be some local omitted shock  $\nu_t$ , that impacts peer firm A's TFP via  $\varepsilon_t$  and impacts target firm B's TFP after a lag via  $\varepsilon_{t-k}$ . This could be a price shock with random adjustment (or)

Additionally, relaxing (21) to allow for weak error autocorrelation between limited  $\varepsilon_{i,t}$  and  $\varepsilon_{i,t-k}$  for limited  $k$  proves a similar problem.

(Firstly, I test for error autocorrelation.)

### 5.4. PANEL VECTOR AUTOREGRESSION

Another mechanism offered by this dynamic setting is to extremely flexibly introduce additional lags of my peer effect term,  $\sum_{j \neq i} w_{i,j,t} z_{j,t}$ .

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_k \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (38)$$

The model now has the form of a panel VAR(p,q) with internal instruments, where  $p$  refers to the autoregressive lags on the dependent variable  $y_{i,t}$  and  $q$  refers to the regressor lags on  $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$  chosen by my specification.

Following above reasoning, lags of  $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$  of a higher order than 2 are exogenous to my reduced-form specification. However,

### 5.5. ALTERNATIVE WEIGHTS ON PEER TFP

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_k \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (39)$$

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APPENDIX 1

TABLE VII  
Dynamic panel: higher-order autoregressive lags

|                      | (1)<br>$\ln(gva_1)$ | (2)<br>$\ln(gva_1)$ | (3)<br>$\ln(gva_1)$ | (4)<br>$\ln(gva_1)$ |
|----------------------|---------------------|---------------------|---------------------|---------------------|
| constant             | 1.371<br>(0.134)*** | 1.446<br>(0.843)    | 0.208<br>(2.578)    | 0.459<br>(2.844)    |
| $\ln(gva_1)_{t-1}$   | 0.005<br>(0.001)*** | 0.019<br>(0.104)    | 0.128<br>(0.295)    | -0.020<br>(0.256)   |
| $\ln(gva_1)_{t-2}$   |                     | 0.002<br>(0.003)    | 0.004<br>(0.101)    | 0.094<br>(0.300)    |
| $\ln(gva_1)_{t-3}$   |                     |                     | -0.001<br>(0.006)   | 0.036<br>(0.162)    |
| $\ln(gva_1)_{t-4}$   |                     |                     |                     | -0.003<br>(0.004)   |
| $\ln(employees)$     | 0.696<br>(0.038)*** | 0.762<br>(0.097)*** | 0.736<br>(0.149)*** | 0.659<br>(0.323)*   |
| $\ln(total\_assets)$ | 0.259<br>(0.041)*** | 0.190<br>(0.090)*   | 0.248<br>(0.121)*   | 0.291<br>(0.305)    |
| $tfp\_wav_1$         | 0.503<br>(0.028)*** | 0.548<br>(0.059)*** | 0.508<br>(0.076)*** | 0.466<br>(0.184)*   |
| Hansen Test          | X<br>(0.417)        | X<br>(0.922)        | X<br>(0.881)        | X<br>(0.826)        |
| AR(1) Test           | ✓<br>(0.000)***     | ✓<br>(0.000)***     | X<br>(0.140)        | X<br>(0.212)        |
| AR(2) Test           | X<br>(0.746)        | X<br>(0.925)        | X<br>(0.802)        | X<br>(0.668)        |
| $i$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   |
| $t$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   |
| Observations         | 713,339             | 603,058             | 511,396             | 433,185             |

TABLE VIII  
Dynamic panel: higher-order regressor lags 1-5

|                      | (1)<br>$\ln(gva_1)$ | (2)<br>$\ln(gva_1)$ | (3)<br>$\ln(gva_1)$ | (4)<br>$\ln(gva_1)$ | (5)<br>$\ln(gva_1)$ |
|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| constant             | 1.124<br>(0.113)*** | 0.956<br>(0.101)*** | 0.743<br>(0.093)*** | 0.584<br>(0.091)*** | 0.543<br>(0.094)*** |
| $\ln(gva_1)_{t-1}$   | 0.002<br>(0.001)*   | 0.003<br>(0.001)*** | 0.005<br>(0.001)*** | 0.005<br>(0.001)*** | 0.005<br>(0.001)*** |
| $tfp\_wav_{1t-1}$    | 0.066<br>(0.006)*** | 0.077<br>(0.007)*** | 0.069<br>(0.007)*** | 0.062<br>(0.007)*** | 0.061<br>(0.008)*** |
| $tfp\_wav_{1t-2}$    |                     | 0.052<br>(0.005)*** | 0.055<br>(0.006)*** | 0.050<br>(0.007)*** | 0.051<br>(0.007)*** |
| $tfp\_wav_{1t-3}$    |                     |                     | 0.043<br>(0.005)*** | 0.048<br>(0.006)*** | 0.049<br>(0.006)*** |
| $tfp\_wav_{1t-4}$    |                     |                     |                     | 0.030<br>(0.005)*** | 0.038<br>(0.006)*** |
| $tfp\_wav_{1t-5}$    |                     |                     |                     |                     | 0.028<br>(0.005)*** |
| $\ln(employees)$     | 0.686<br>(0.036)*** | 0.695<br>(0.038)*** | 0.670<br>(0.040)*** | 0.644<br>(0.044)*** | 0.665<br>(0.052)*** |
| $\ln(total\_assets)$ | 0.270<br>(0.040)*** | 0.260<br>(0.042)*** | 0.287<br>(0.045)*** | 0.317<br>(0.049)*** | 0.294<br>(0.057)*** |
| $tfp\_wav_1$         | 0.508<br>(0.028)*** | 0.514<br>(0.029)*** | 0.492<br>(0.030)*** | 0.469<br>(0.033)*** | 0.480<br>(0.038)*** |
| Hansen Test          | X<br>(0.492)        | X<br>(0.968)        | X<br>(0.453)        | X<br>(0.084)*       | X<br>(0.080)*       |
| AR(1) Test           | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     |
| AR(2) Test           | X<br>(0.279)        | X<br>(0.612)        | X<br>(0.801)        | X<br>(0.854)        | X<br>(0.729)        |
| $i$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   |
| $t$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   |
| Observations         | 713,339             | 603,058             | 511,396             | 433,185             | 365,908             |

TABLE IX  
Dynamic panel: higher-order regressor lags 6-10

|                      | (1)<br>$\ln(gva_1)$ | (2)<br>$\ln(gva_1)$ | (3)<br>$\ln(gva_1)$ | (4)<br>$\ln(gva_1)$ | (5)<br>$\ln(gva_1)$ |
|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| constant             | 0.421<br>(0.082)*** | 0.388<br>(0.081)*** | 0.270<br>(0.077)*** | 0.199<br>(0.083)*   | 0.116<br>(0.093)    |
| $\ln(gva_1)_{t-1}$   | 0.004<br>(0.001)*** | 0.004<br>(0.001)**  | 0.003<br>(0.001)*   | 0.002<br>(0.002)    | 0.002<br>(0.002)    |
| $tfp\_wav_{1t-1}$    | 0.055<br>(0.008)*** | 0.056<br>(0.008)*** | 0.047<br>(0.008)*** | 0.042<br>(0.009)*** | 0.036<br>(0.011)**  |
| $tfp\_wav_{1t-2}$    | 0.045<br>(0.007)*** | 0.045<br>(0.007)*** | 0.040<br>(0.007)*** | 0.037<br>(0.007)*** | 0.035<br>(0.009)*** |
| $tfp\_wav_{1t-3}$    | 0.046<br>(0.006)*** | 0.046<br>(0.006)*** | 0.039<br>(0.006)*** | 0.033<br>(0.007)*** | 0.029<br>(0.008)*** |
| $tfp\_wav_{1t-4}$    | 0.034<br>(0.006)*** | 0.031<br>(0.006)*** | 0.024<br>(0.006)*** | 0.019<br>(0.007)**  | 0.017<br>(0.008)*   |
| $tfp\_wav_{1t-5}$    | 0.029<br>(0.005)*** | 0.030<br>(0.006)*** | 0.023<br>(0.006)*** | 0.019<br>(0.006)**  | 0.017<br>(0.006)**  |
| $tfp\_wav_{1t-6}$    | 0.018<br>(0.004)*** | 0.024<br>(0.005)*** | 0.021<br>(0.005)*** | 0.019<br>(0.005)*** | 0.017<br>(0.006)**  |
| $tfp\_wav_{1t-7}$    |                     | 0.019<br>(0.005)*** | 0.017<br>(0.005)*** | 0.015<br>(0.006)**  | 0.011<br>(0.007)    |
| $tfp\_wav_{1t-8}$    |                     |                     | 0.013<br>(0.004)**  | 0.016<br>(0.005)**  | 0.017<br>(0.006)**  |
| $tfp\_wav_{1t-9}$    |                     |                     |                     | 0.010<br>(0.005)*   | 0.013<br>(0.005)*   |
| $\ln(employees)$     | 0.628<br>(0.049)*** | 0.636<br>(0.053)*** | 0.582<br>(0.055)*** | 0.548<br>(0.061)*** | 0.523<br>(0.073)*** |
| $tfp\_wav_{1t-10}$   |                     |                     |                     |                     | 0.016<br>(0.006)**  |
| $\ln(total\_assets)$ | 0.334<br>(0.054)*** | 0.325<br>(0.058)*** | 0.384<br>(0.061)*** | 0.421<br>(0.067)*** | 0.448<br>(0.080)*** |
| $tfp\_wav_1$         | 0.452<br>(0.036)*** | 0.457<br>(0.038)*** | 0.420<br>(0.040)*** | 0.397<br>(0.044)*** | 0.380<br>(0.052)*** |
| Hansen Test          | X<br>(0.097)*       | X<br>(0.122)        | X<br>(0.450)        | X<br>(0.637)        | X<br>(0.435)        |
| AR(1) Test           | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     | ✓<br>(0.000)***     |
| AR(2) Test           | X<br>(0.924)        | X<br>(0.962)        | X<br>(0.568)        | X<br>(0.507)        | X<br>(0.450)        |
| $i$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   |
| $t$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   |
| Observations         | 307,394             | 256,016             | 210,798             | 171,125             | 136,527             |

TABLE X  
Dynamic panel: higher-order VAR(2,4) structures

|                      | (1)<br>$\ln(gva_1)$ | (2)<br>$\ln(gva_1)$ | (3)<br>$\ln(gva_1)$ | (4)<br>$\ln(gva_1)$ | (5)<br>$\ln(gva_1)$ | (6)<br>$\ln(gva_1)$ |
|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| constant             | 1.164<br>(0.573)*   | 0.498<br>(1.561)    | 0.398<br>(1.667)    | 1.494<br>(0.428)*** | 0.333<br>(1.469)    | 1.351<br>(0.307)*** |
| $\ln(gva_1)_{t-1}$   | -0.011<br>(0.117)   | 0.081<br>(0.280)    | -0.028<br>(0.251)   | -0.129<br>(0.081)   | 0.087<br>(0.276)    | -0.125<br>(0.063)*  |
| $\ln(gva_1)_{t-2}$   | -0.000<br>(0.003)   | 0.006<br>(0.111)    | 0.076<br>(0.385)    | 0.004<br>(0.002)    | -0.001<br>(0.111)   | 0.006<br>(0.002)**  |
| $\ln(gva_1)_{t-3}$   |                     | -0.001<br>(0.005)   | 0.034<br>(0.156)    |                     | -0.002<br>(0.004)   |                     |
| $\ln(gva_1)_{t-4}$   |                     |                     | -0.002<br>(0.004)   |                     |                     |                     |
| $tfp\_wav_1$         | 0.551<br>(0.052)*** | 0.504<br>(0.078)*** | 0.475<br>(0.201)*   | 0.525<br>(0.049)*** | 0.488<br>(0.073)*** | 0.524<br>(0.050)*** |
| $tfp\_wav_{1t-1}$    | 0.099<br>(0.102)    | 0.005<br>(0.255)    | 0.082<br>(0.217)    | 0.196<br>(0.073)**  | -0.005<br>(0.249)   | 0.189<br>(0.055)*** |
| $tfp\_wav_{1t-2}$    | 0.060<br>(0.011)*** | 0.036<br>(0.099)    | -0.019<br>(0.342)   | 0.072<br>(0.012)*** | 0.046<br>(0.102)    | 0.070<br>(0.010)*** |
| $tfp\_wav_{1t-3}$    |                     |                     |                     | 0.056<br>(0.009)*** | 0.039<br>(0.020)    | 0.065<br>(0.008)*** |
| $tfp\_wav_{1t-4}$    |                     |                     |                     |                     |                     | 0.042<br>(0.006)*** |
| $\ln(employees)$     | 0.737<br>(0.099)*** | 0.720<br>(0.146)*** | 0.643<br>(0.382)    | 0.650<br>(0.075)*** | 0.705<br>(0.148)*** | 0.657<br>(0.079)*** |
| $\ln(total\_assets)$ | 0.209<br>(0.088)*   | 0.253<br>(0.119)*   | 0.307<br>(0.367)    | 0.272<br>(0.074)*** | 0.271<br>(0.119)*   | 0.266<br>(0.079)*** |
| Hansen Test          | X<br>(0.971)        | X<br>(0.855)        | X<br>(0.849)        | X<br>(0.999)        | X<br>(0.851)        | X<br>(0.232)        |
| AR(1) Test           | ✓<br>(0.002)***     | X<br>(0.124)        | X<br>(0.163)        | ✓<br>(0.000)***     | X<br>(0.111)        | ✓<br>(0.000)***     |
| AR(2) Test           | X<br>(0.889)        | X<br>(0.878)        | X<br>(0.775)        | X<br>(0.092)*       | X<br>(0.837)        | ✓<br>(0.030)**      |
| $i$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   |
| $t$ fixed effects    | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   |
| Observations         | 603,058             | 511,396             | 433,185             | 511,396             | 511,396             | 433,185             |

## APPENDIX 2

## APPENDIX 3